

Data Mining For Decision Making

Linear and Logistic Regression

Prof. Varun Dutt

School of Computing and Electrical Engineering
Indian Institute of Technology Mandi, India



Scaling the Heights

What is Regression?

Concept:

- Regression = A **predictive modeling technique**.
- Helps us understand the **relationship** between:
 - **Independent variable(s) (X)** → inputs/predictors
 - **Dependent variable (Y)** → output/response

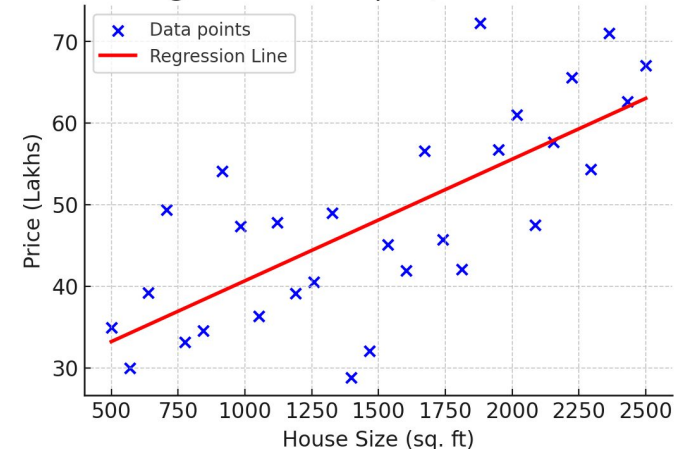
Linear Regression:

- Fits a **straight line** to data.
- Equation:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Where:
 - β_0 : intercept
 - β_1 : slope (rate of change)
 - ε : error

Linear Regression Example (House Size vs Price)



Scatter plot of **house size vs. price**, with a red regression line showing how the model predicts Y

Simple Linear Regression Equation

Concept:

- Linear regression models the **relationship** between one predictor X and outcome Y .
- Equation:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Where:
 - β_0 = Intercept (value of Y when $X = 0$)
 - β_1 = Slope (change in Y for each 1 unit increase in X)
 - ε = Error (residual difference between observed and predicted Y)

Example (House Prices):

- Suppose:

$$\text{Price} = 20 + 0.02 \times \text{Size}$$

- Interpretation:
 - Each extra **100 sq. ft** increases house price by **~2 lakhs**.
 - A 0 sq. ft "theoretical" house would cost **20 lakhs** (intercept).

Visual Example with Residuals

Concept:

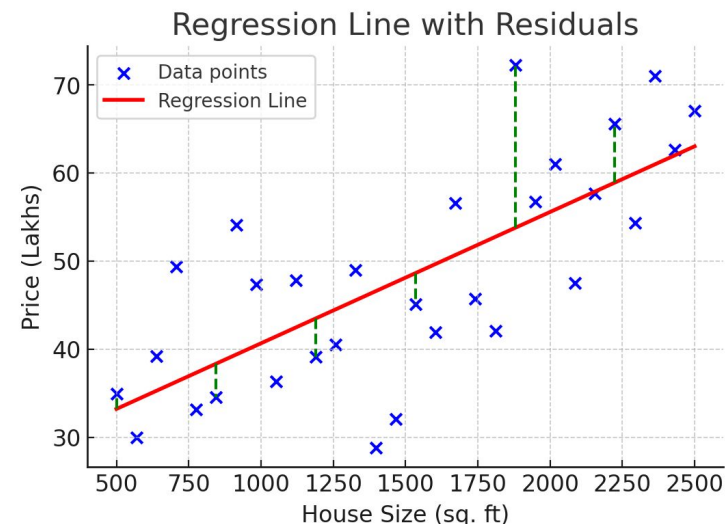
- The **red line** is the regression line (predicted values).
- The **blue points** are actual observed values.
- **Residuals (errors)** = Vertical distance between each point and the line.

Geometric View:

- $\text{Residual} = \text{Actual } Y - \text{Predicted } Y$
- The best line minimizes the **sum of squared residuals (SSE)**.

Example (House Price Prediction):

- Actual = ₹55 lakh
- Predicted = ₹53 lakh
- Residual = ₹2 lakh



Assumptions of Linear Regression

For regression results to be valid, these assumptions must hold:

1. Linearity

- The relationship between X and Y is linear.
- Example: House size $\leftrightarrow$ Price (approx. straight line).

2. Independence

- Observations are independent of each other.
- Example: One house's price doesn't influence another's.

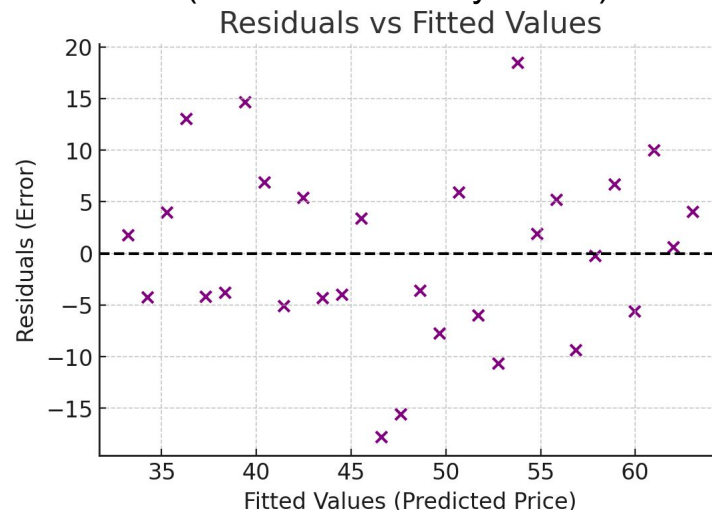
3. Homoscedasticity (Equal variance of errors)

- Spread of residuals is constant across values of X.
- Example: Prediction errors should not systematically increase for larger houses.

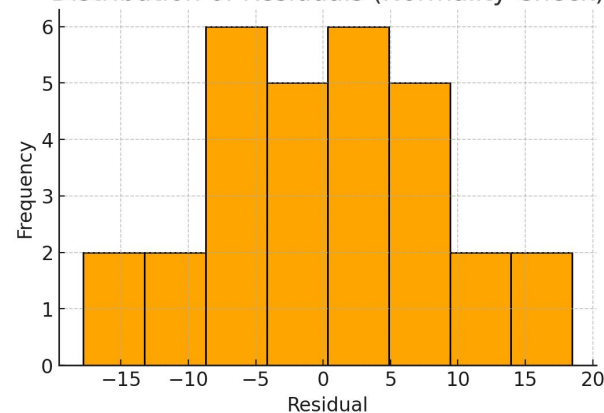
4. Normality of Errors

- Residuals (errors) should follow a normal distribution.
- Important for valid hypothesis tests (t-test, F-test).

(Homoscedasticity check)



Distribution of Residuals (Normality Check)



Evaluating Linear Regression: R^2

R^2 : Coefficient of Determination

- Measures **how much variance in Y is explained** by the regression model.
- Formula:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

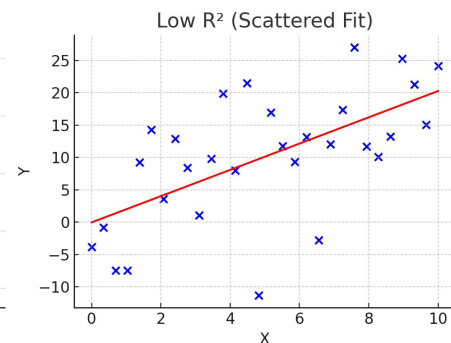
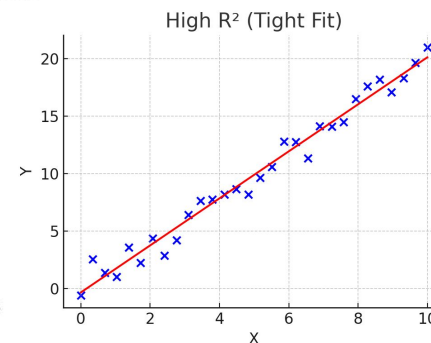
- SS_{res} = sum of squared residuals
- SS_{tot} = total variance in Y

Interpretation:

- $R^2 = 0 \rightarrow$ model explains nothing (no better than mean).
- $R^2 = 1 \rightarrow$ model perfectly explains all variance.

Example (House Price Prediction):

- $R^2 = 0.82$
- Means: 82% of variation in house price is explained by house size.



Computing SSres and SStot

Formula Recap:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

Step 1: Total Sum of Squares (SS_{tot})

- Measures the **total variation** in Y (how much Y varies around its mean).

$$SS_{tot} = \sum_{i=1}^n (y_i - \bar{y})^2$$

- y_i = actual values
- $\bar{y}$ = mean of Y

Interpretation: How spread out the actual Y values are.

Step 2: Residual Sum of Squares (SS_{res})

- Measures the **unexplained variation** (errors between actual and predicted values).

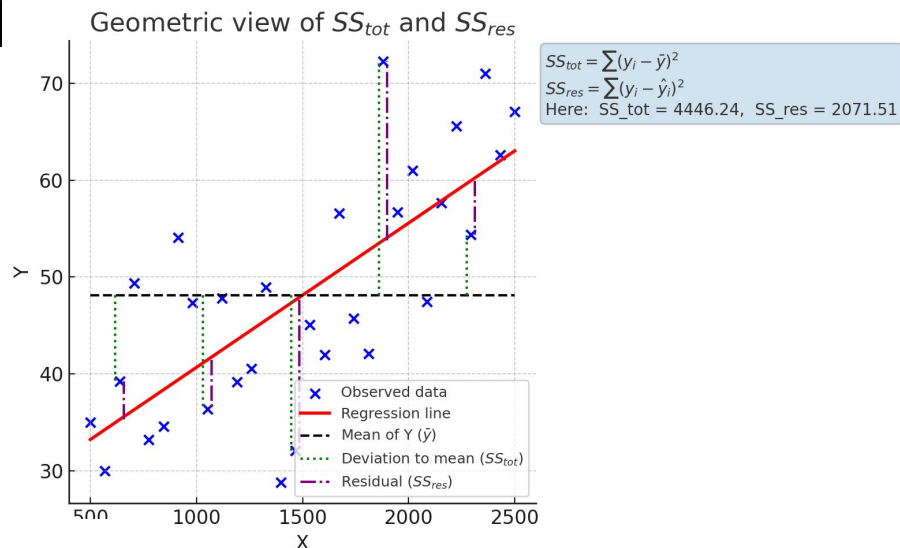
$$SS_{res} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- y_i = actual values
- $\hat{y}_i$ = predicted values from regression

Interpretation: How far the points are from the regression line.

Example (House Prices):

- Actual price: 55, Predicted: 53 $\rightarrow$ Residual = 2
- Contribution to SS_{res} : $2^2 = 4$



Multiple Linear Regression

Concept:

- Extends simple regression to **more than one predictor**.
- Equation:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \varepsilon$$

- Each β_i measures effect of predictor X_i , holding others constant.

Multiple Linear Regression: Salary vs Education & Experience

Example (Salary Prediction):

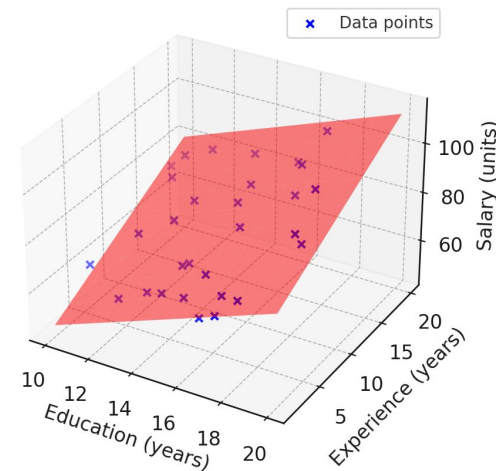
- Predict **Salary (Y)** from:
 - Education level (X_1)
 - Age (X_2)
 - Years of experience (X_3)

Equation might look like:

$$\text{Salary} = 15 + 2.5(\text{Education}) + 0.3(\text{Age}) + 1.2(\text{Experience})$$

Interpretation:

- Each **additional year of experience** increases salary by 1.2 units, if education and age remain constant.
- Coefficients show contribution of each factor separately.



Evaluating Linear Regression: F-ratio

Concept:

- Even if one predictor seems useful, we need to test if the **overall regression model** explains significant variation in Y.
- This is done with the **F-test**.

Formula:

$$F = \frac{MS_{regression}}{MS_{residual}}$$

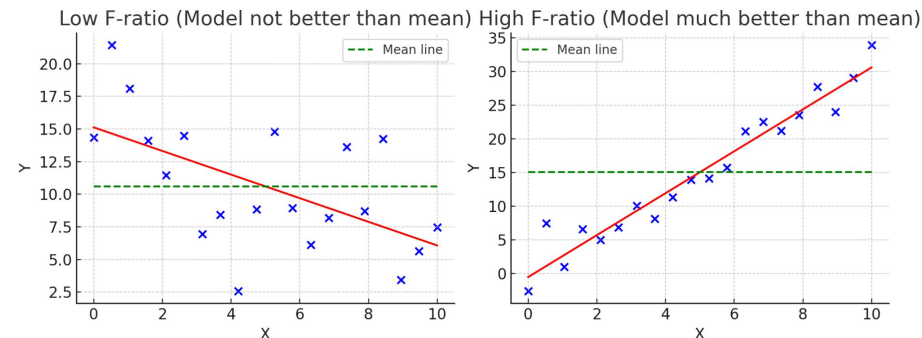
- $MS_{regression}$: Mean squares due to regression
- $MS_{residual}$: Mean squares due to residuals

Interpretation:

- Large **F** → model explains a lot more variance than error.
- Compare F against critical value (or p-value).

Example:

- Salary model with 3 predictors
- $F = 25.3$, $p < 0.001$ → model is statistically significant.



Shows two scenarios:

- **Small F** → Regression line not much better than mean line.
- **Large F** → Regression line clearly explains variation.

Understanding the F-ratio

Step 1: Compute Sum of Squares

- Regression Sum of Squares (SSR)

$$SSR = \sum (\hat{y}_i - \bar{y})^2$$

Variation explained by the regression model.

- Residual Sum of Squares (SSE)

$$SSE = \sum (y_i - \hat{y}_i)^2$$

Variation not explained (errors).

Step 2: Convert to Mean Squares

- Mean Square Regression (MS_{regression})

$$MS_{regression} = \frac{SSR}{df_{regression}}$$

- Mean Square Residual (MS_{residual})

$$MS_{residual} = \frac{SSE}{df_{residual}}$$

- Degrees of freedom:

- $df_{regression} = p$ (number of predictors)
- $df_{residual} = n - p - 1$

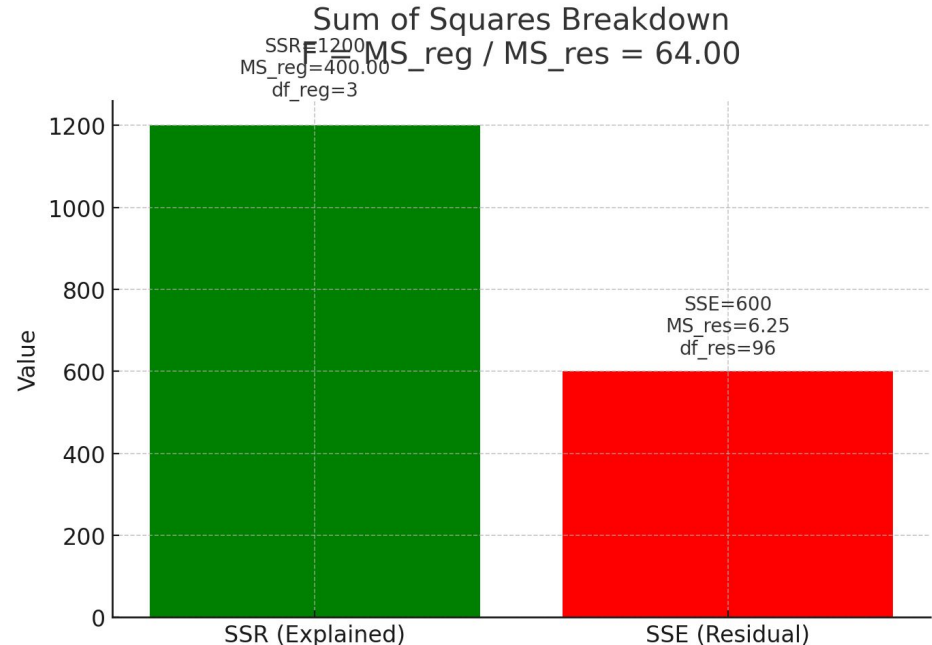
Step 3: Compute F

- $$F = \frac{MS_{regression}}{MS_{residual}}$$

- Large F → model explains much more variance than error → model is significant.

Example

- Salary model with 3 predictors
- $SSR = 1200, df_{reg} = 3 \Rightarrow MS_{reg} = 400$
- $SSE = 600, df_{res} = 96 \Rightarrow MS_{res} = 6.25$
- $F = 400/6.25 = 64 \rightarrow$ highly significant.



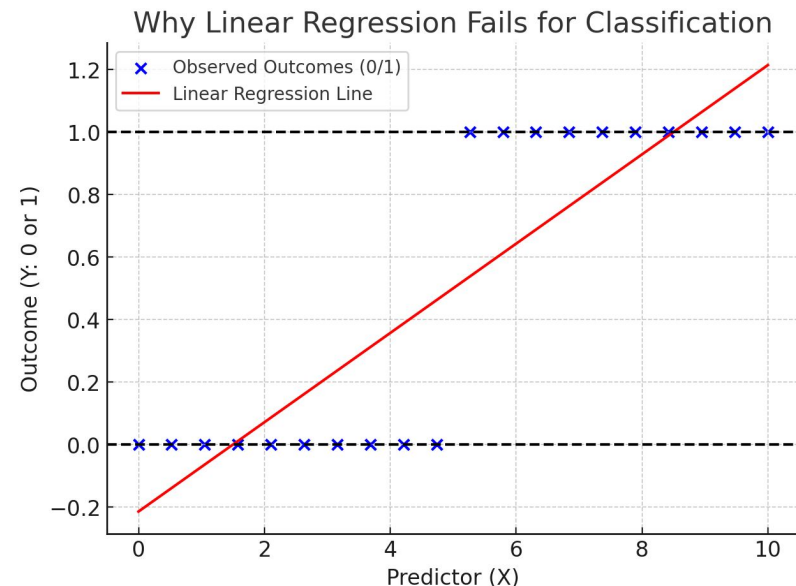
Limitations of Linear Regression

Why Linear Regression Fails for Classification:

- Works well when **Y is continuous**.
- Fails when **Y is categorical (Yes/No, 0/1)**:
 - Predictions can be **negative** or **greater than 1** → not valid probabilities.
 - Line doesn't capture the S-shaped probability trend.

Example (Disease Prediction):

- Predicting whether a patient has a disease (Yes/No).
- Linear regression might give:
 - $P(\text{disease}) = -0.3 \rightarrow$ meaningless.
 - $P(\text{disease}) = 1.2 \rightarrow$ impossible.



Introduction to Logistic Regression

Concept:

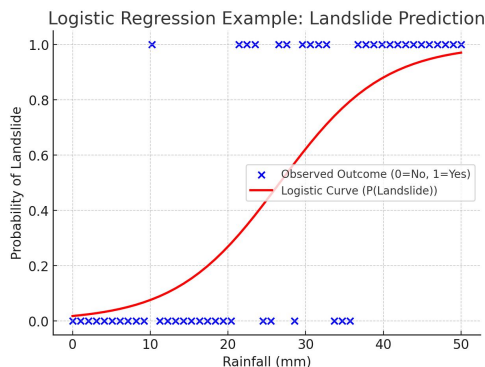
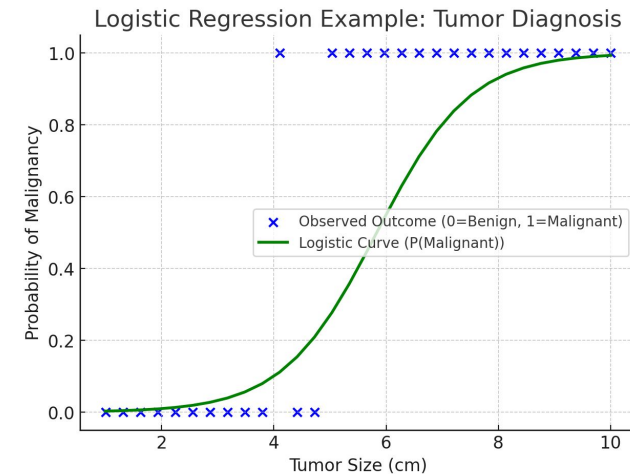
- Logistic regression is used when **Y is categorical** (binary: 0/1, or multinomial).
- Predicts **probability of event occurrence**.
- Uses **sigmoid (S-shaped) curve** to map any input to [0,1].

Equation:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Example (Tumor Diagnosis):

- Predict whether tumor is **malignant (1)** or **benign (0)**.
- As tumor size increases, probability of malignancy rises smoothly between 0 and 1.



Blue dots = observed landslide outcomes (0 = No, 1 = Yes).

Red S-curve = Logistic regression prediction of landslide probability.

Nicely shows how probability of landslide increases with rainfall while staying within **0–1 bounds**.

Blue dots = observed tumor outcomes (Benign = 0, Malignant = 1).

Green S-curve = Logistic regression prediction of malignancy probability.

Shows how probability smoothly increases with tumor size, always staying between **0 and 1**.

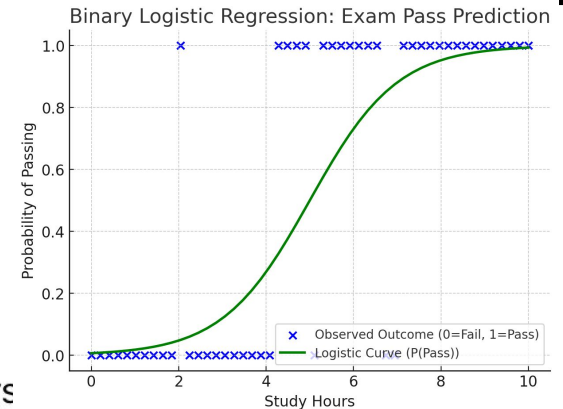
Binary vs. Multinomial Logistic Regression

Binary Logistic Regression:

- Outcome has **two categories** (Yes/No, 0/1).
- Equation predicts probability of one outcome vs. the other.

Example:

- Predicting if a student **passes (1)** or **fails (0)** an exam.
- Logistic curve shows probability of passing increasing with study hours

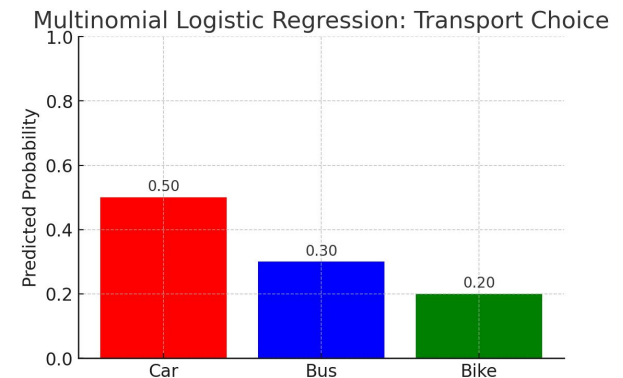


Multinomial Logistic Regression:

- Outcome has **more than two categories**.
- Compares each category to a **baseline category**.

Example:

- Predicting **transport choice** (Car, Bus, Bike).
- Model gives probability of each transport option given income, distance, etc.



Evaluating Logistic Regression: Log-likelihood

Why R^2 Doesn't Apply:

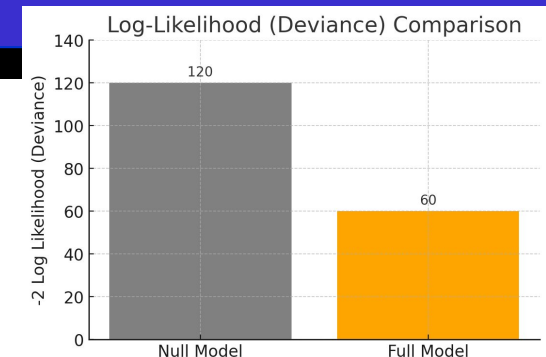
- Logistic regression predicts probabilities, not continuous Y.
- Variance explained (R^2) doesn't make sense.

Alternative Measure: Log-likelihood

- Indicates how well model explains data.
- Higher log-likelihood = better model fit.
- Often reported as **-2 Log Likelihood (Deviance)** → lower is better.

Example:

- Null model (no predictors): $-2LL = 120$
- Model with predictors: $-2LL = 60$
- Reduction shows model fit improved significantly.



- Bar chart comparing **-2 Log-Likelihood (Deviance)** for:
 - **Null Model (120)** – poor fit.
 - **Full Model (60)** – better fit.

This clearly shows how adding predictors improves model fit (lower deviance).

What is Log-Likelihood (LL)?

- Logistic regression is fitted using **Maximum Likelihood Estimation (MLE)**.
- Likelihood (L) = probability of observing the actual data, given model parameters.
- Log-Likelihood (LL) = natural log of L.
 - Used because logs turn products into sums (easier math).

$$LL = \sum_{i=1}^n \left(y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i) \right)$$

- y_i = actual outcome (0 or 1)
- p_i = predicted probability from the logistic model

Pseudo R² Measures (Logistic Regression)

1. Cox & Snell R²

$$R_{CS}^2 = 1 - \left(\frac{L_0}{L_M} \right)^{\frac{2}{n}}$$

- L_0 : Likelihood of null model (no predictors)
- L_M : Likelihood of fitted model
- n : Sample size
- Problem: Can never reach 1.

2. Nagelkerke R²

$$R_N^2 = \frac{R_{CS}^2}{1 - L_0^{\frac{2}{n}}}$$

- Scales Cox & Snell so that max = 1.

3. McFadden R²

$$R_{McF}^2 = 1 - \frac{\ln L_M}{\ln L_0}$$

- Compares log-likelihoods directly.
- Rule of thumb:
 - 0.2 – 0.4 → excellent fit.

Example Calculation (Tumor Classification)

- Null Model log-likelihood: –120
- Full Model log-likelihood: –80
- McFadden's R² = $1 - (-80 / -120) = 0.33 \rightarrow$ good fit.

Classification Metrics (Logistic Regression)

Evaluating Predictions:

1. Confusion Matrix

- Compares predicted vs. actual outcomes.
- Example (Tumor Classification):

	Predicted: Benign	Predicted: Malignant
Actual: Benign	True Negative (TN)	False Positive (FP)
Actual: Malignant	False Negative (FN)	True Positive (TP)

2. Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

3. ROC Curve & AUC

- Plots True Positive Rate vs. False Positive Rate.
- AUC = probability classifier ranks a random positive > negative.
- AUC closer to 1 = **better performance**.

Example

Patient	Actual (y)	Predicted Prob (p)
1	0	0.10
2	0	0.15
3	1	0.80
4	0	0.30
5	1	0.60
6	1	0.75
7	0	0.20
8	1	0.85
9	0	0.40
10	1	0.90

Example

Step 1 – Confusion Matrix (threshold = 0.5)

Rule: If $p \geq 0.5 \rightarrow$ predict 1 (Malignant), else 0 (Benign).

Predictions:

- Patients 3,5,6,8,10 $\rightarrow$ predicted 1
- Patients 1,2,4,7,9 $\rightarrow$ predicted 0

Compare with actual outcomes:

- **True Positives (TP):** 5 (patients 3,5,6,8,10)
- **True Negatives (TN):** 4 (patients 1,2,4,7)
- **False Positives (FP):** 0
- **False Negatives (FN):** 1 (patient 9)

Confusion Matrix

	Pred 0	Pred 1
Actual 0	TN=4	FP=0
Actual 1	FN=1	TP=5

Step 2 – Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} = \frac{5 + 4}{5 + 4 + 0 + 1} = \frac{9}{10} = 0.90$$

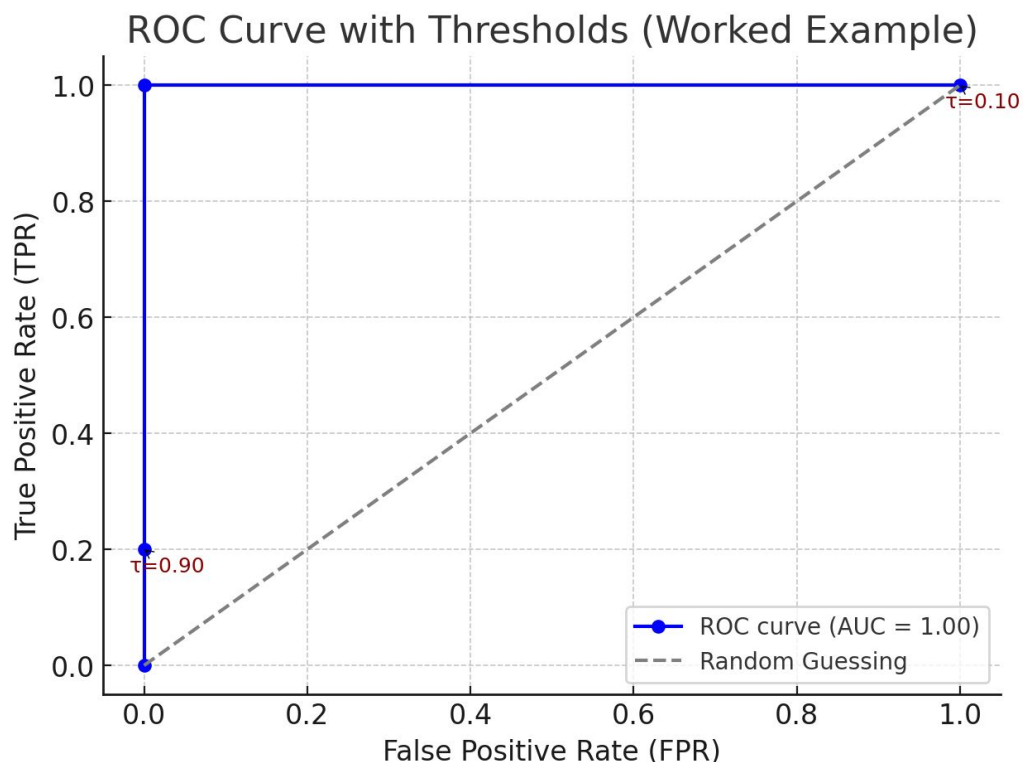
Step 3 – ROC Curve and AUC

- Vary threshold τ from 1 $\rightarrow$ 0.
- At each τ , compute TPR (Recall) and FPR.
- Plot ROC: TPR vs FPR.

Interpretation:

- As τ decreases, more positives are predicted $\rightarrow$ TPR $\uparrow$ but FPR also $\uparrow$.
- AUC = probability model ranks a random malignant $>$ benign.

Example



- **Blue curve** = ROC performance of the logistic model.
- **Dashed gray line** = random classifier baseline.
- **Red threshold labels ($\tau=0.90, 0.70, 0.50, 0.30, 0.10$)** are annotated with arrows so students can clearly see how TPR/FPR change as the cutoff moves.
- **AUC ≈ 1.00** shows excellent classifier quality.

AUC:

1. **Compute ROC points: (FPR, TPR) for all thresholds.**
2. **Connect the points.**
3. **Use trapezoidal rule to compute area.**

Odds Ratio in Logistic Regression

Odds

- Odds = Probability of event ÷ Probability of no event
- Example: If $P(\text{landslide}) = 0.8 \rightarrow \text{Odds} = 0.8 / 0.2 = 4$
(4-to-1 odds in favor of landslide occurring)

Odds Ratio (OR)

- Change in odds for a **1-unit increase** in predictor.
- Formula:

$$OR = e^{\beta_1}$$

- If $OR = 1 \rightarrow$ No effect
- If $OR > 1 \rightarrow$ Predictor increases odds
- If $OR < 1 \rightarrow$ Predictor decreases odds

Example (Rainfall & Landslide)

- Logistic regression coefficient $\beta_1 = 0.08$
- Odds Ratio = $e^{0.08} \approx 1.08$
- **Interpretation:** Each additional 1 mm rainfall increases odds of landslide by **8%**.

Example: Landslide prediction

- $\beta_1 = 0.08$ for rainfall
- Odds Ratio = $e^{0.08} = 1.08$
- Interpretation: Each additional mm of rainfall **increases odds of landslide by 8%**

Summary

✓ Linear Regression

- Predicts continuous outcomes (house price, salary).
- Evaluated with R^2 and F-test.
- Assumptions: linearity, independence, homoscedasticity, normality.

✓ Logistic Regression

- Predicts categorical outcomes (Yes/No, class labels).
- Evaluated with log-likelihood, pseudo R^2 , confusion matrix, ROC/AUC.
- Odds Ratios help interpret predictor influence.

Key Takeaway:

- Regression is a **fundamental predictive tool**.
- Choose **Linear Regression** for continuous outcomes,
- and **Logistic Regression** for categorical outcomes.